

NAFIZ IMTIAZ RAFIN

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OBJECTIVE

Motivated to pursue a PhD in Computer Science with a research focus on machine learning, deep learning, computer vision, AI, and computational biology. My interests extend to reinforcement learning, federated learning, and explainable AI, aiming to develop innovative solutions for challenges in healthcare, urban systems, and security. I aspire to advance cutting-edge research in AI while contributing to academia through mentorship, collaboration, and impactful scholarly work.

EXPERIENCE

- BRAC University**  October 2024 - Present
Adjunct Lecturer Dhaka, Bangladesh
 - **Course Instruction:** Teaching undergraduate courses, including Data Structures, Algorithms, Digital Logic Design, Operating Systems, Data Science, and Artificial Intelligence.
 - **Curriculum Development:** Developing course materials, assignments, and exams to enhance students' understanding of fundamental and advanced computer science concepts.
 - **Student Mentorship:** Providing academic guidance and mentorship to students, helping them with both course material and career and research advice.
- BRAC University**  September 2023 - April 2025
Research Assistant Dhaka, Bangladesh
 - Conducting research on blockchain technology and reinforcement learning algorithms, Data analysis, computer vision, Deep Learning, Machine Learning, etc.
 - Published findings on optimal block formation in Bitcoin blockchain networks using Recurrent Proximal Policy Optimization, policy-making depending on machine learning in fragile families, neural networks and computer vision, cyber security, and network intrusion detection and optimization.
 - Presented research at international conferences, including ICCA 2024, ICEEICT 2024, and ICCIT 2024
 - Guiding prospective undergraduate thesis students in developing novel and impactful research.
- BRAC University**  August 2021 - January 2023
Student Tutor Dhaka, Bangladesh
 - Tutored undergraduate students in data structures, algorithms, and introductory programming courses.
 - Developed teaching materials and assisted in lab sessions, enhancing student understanding of complex concepts.
- Pencils Education** March 2021 - July 2022
Co-Founder and Instructor Dhaka, Bangladesh
 - Taught Physics, Computer Science, and ICT courses, providing tailored instruction to diverse student groups.

EDUCATION

- BRAC University** January 2019 - January 2023
BSc in Computer Science and Engineering Dhaka, Bangladesh
 - CGPA: 3.87/4.00 (**Highest Distinction**)
- Alfred International School and College** June 2018
Advanced Level (A Level) - EDEXCEL Dhaka, Bangladesh
 - GPA: 5.00/5.00
- Alfred International School and College** June 2015
Ordinary Level (O Level) - EDEXCEL Dhaka, Bangladesh
 - GPA: 5.00/5.00

PROJECTS

• Federated Learning-Based Tuberculosis Classification from X-Ray Images

Ongoing

Tools: Federated Learning, TensorFlow, Python

- Focused on developing federated learning algorithms to enable privacy-preserving medical image classification across multiple institutions. This project aims to improve tuberculosis diagnosis while preserving patient data privacy.

• Neurosymbolic AI-Based Project for Explainable AI Models

Ongoing

Tools: Neural Networks, Symbolic Reasoning

- Combining neural networks with symbolic reasoning to develop explainable AI models. The project focuses on providing transparency in AI decision-making, particularly in critical areas like healthcare.

• GNN and RL for Viral Protein Structural Disorder Fold Identification and Drug Discovery

Ongoing

Tools: Graph Neural Networks, Reinforcement Learning

- In this project, I am utilizing attention-based Graph Neural Networks and Reinforcement Learning to identify optimized protein folds with lower energy levels in viral protein RNA, contributing to protein structural analysis, resulting in faster and more efficient drug discovery.

RESEARCH WORKS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [J.1] ROBB: Recurrent Proximal Policy Optimization Reinforcement Learning for Optimal Block Formation in Bitcoin Blockchain Network. *IEEE Access Volume 12* <https://ieeexplore.ieee.org/document/10445192>.
- [J.2] Exploring Childhood Disabilities in Fragile Families: Machine Learning Insights for Informed Policy Interventions. *Journal of Disability Research*, April 2024
<https://www.scienceopen.com/hosted-document?doi=10.57197/JDR-2024-0032>.
- [J.3] A Method of Hybridizing and Cascading Several Learning Phases to Detect Brain Stroke in MRI and CT Scan Images. *International Journal of Computer Engineering and Sciences Research*, VOL. 06, NO. 01, September-October 2024, Pages 01–17
- [S.J.1] Integrating Self-Attention and Ensemble Models to Forecast Developmental Trajectories for Children in Fragile Family Dynamics *Submitted to Information Sciences, Elsevier*.
- [S.J.2] Ada-KDRL: Adaptive Knowledge Distillation with Reinforcement Learning for Efficient Multitemporal Satellite Building Detection *Submitted to ISPRSS, Elsevier*.
- [S.J.3] Aggregation over Architecture: A Training-Free Audit of Resolution Robustness in 3D Shape Representations *Submitted to PeerJ Computer Science*.
- [S.J.4] Label-Efficient Ocular Toxoplasmosis Diagnosis via Foundation Model Adaptation: A Systematic Study of Fine-Tuning Strategies, Augmentation, and Class Imbalance Handling *Submitted to Scientific Reports, Nature*
- [S.J.5] MAMT: Masked Autoencoder and MAMBA-TemporalGan Based Self-Supervised Segmentation of SAR Imagery for Riverbank Erosion Detection *Submitted to IEEE GRSS*.
- [S.J.6] Curriculum Modality Dropout and Self-Consistency for Robust Optical–SAR Flood Mapping with Missing Sensors *Submitted to Nature Scientific Reports*.
- [C.1] Enhancing Intrusion Detection: A Robust Ensemble Learning Approach with Explainable AI and Strategic Fused Voting-Stacking Mechanism. *ICCA '24: Proceedings of the 3rd International Conference on Computing Advancements Pages 1050 - 1057* <https://doi.org/10.1145/3723178.3723317>.
- [C.2] Transforming Precision: A Comparative Analysis of Vision Transformers, CNNs and Traditional ML for Knee Osteoarthritis Severity Diagnosis. *6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT 2024)* <https://ieeexplore.ieee.org/document/10534528>.
- [C.3] Smartphone Sensor-based Human Activity Recognition Using LSTM Networks: Development and Implementation in Android Applications. *2024 27th International Conference on Computer and Information Technology (ICCIT)* <https://ieeexplore.ieee.org/abstract/document/11021865>.

- [C.4] BDConvSGNet: A Lightweight Neural Network for Isolated Bangla Digit Sound Classification. 2024 27th International Conference on Computer and Information Technology (ICCIT) <https://ieeexplore.ieee.org/abstract/document/11022527>.
- [C.5] Peer-Guided Optimization: Incorporating Collaborative Learning into Stochastic Optimization in Machine Learning 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE) <https://ieeexplore.ieee.org/document/11013057>.
- [C.6] An Integrated Approach to Forecast Bitcoin Price Incorporating Economic Factors, Strategic Commodities, and X Tweets 2025 28th International Conference on Computer and Information Technology (ICCIT) <https://ieeexplore.ieee.org/abstract/document/11491428>.
- [C.7] Optimal Transport Theory Based GAN for Medical Image Augmentation and Classification. 2025 28th International Conference on Computer and Information Technology (ICCIT) <https://ieeexplore.ieee.org/abstract/document/11491184/>.
- [C.8] Optimizing Building Energy Consumption with Genetic Algorithms and Explainable AI 2025 28th International Conference on Computer and Information Technology (ICCIT) <https://ieeexplore.ieee.org/abstract/document/11491401>.
- [C.9] Minimizing the Effect of Sleep Deprivation in the Forward-Forward Algorithm. ICPR 2026 https://link.springer.com/chapter/10.1007/978-3-032-31933-3_40.
- [S.C.1] SLA and Cost-aware Cloud Service Provisioning through Matching Game and TOPSIS. Submitted to ICCIT 2026.
- [S.C.2] High-Fidelity Turbulence Modeling: Enhancing Accuracy Through Physics-Informed, Tensor-Basis Neural Networks Submitted to ICCIT 2026.
- [S.C.3] Cyber Threat Classification Using Natural Language Processing with BERT-Based Deep Learning Submitted to ICCIT 2026.

SKILLS

- **Programming Languages:** JAVA, Python, C, C++, R, HTML, CSS, PHP
- **Web Technologies:** Laravel, Django
- **Database Systems:** MySQL
- **Data Science & Machine Learning:** Matplotlib, Pandas, Numpy, Scikit-learn, Seaborn, Keras, TensorFlow, PyTorch, OpenCV, Explainable AI (Lime, SHAP)
- **Cloud Technologies:** AWS, Google Cloud Platform (GCP)
- **DevOps & Version Control:** Git, GitHub, Docker, Jenkins
- **Specialized Area:** Bioinformatics and Computational Biology, Reinforcement Learning, Explainable AI, Neural Networks, Remote Sensing
- **Mathematical & Statistical Tools:** R, MATLAB, StatsModels
- **Other Tools & Technologies:** IntelliJ, NetBeans, Eclipse, VS Code, Jupyter Notebook
- **Research Skills:** Literature Review, Experimental Design, Data Analysis, Statistical Modeling, Technical Writing, Presentation Skills

REFERENCES

1. Dr. Md. Golam Rabiul Alam

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Undergraduate Thesis Supervisor

2. Dr. Ashraful Alam

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Research Advisor, CVIS Lab

3. **Mr. Md. Tanzim Reza**

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Course Instructor

4. **Dr. Mohammad Mehedi Hassan**

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Project Supervisor